

Stylometric analysis of AI-generated texts: a comparative study of ChatGPT and DeepSeek

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ABSTRACT

This study examined the readability and stylometric characteristics of AI-generated texts (AIGTs) produced by ChatGPT and DeepSeek. Specifically, it examines the extent to which AIGTs differ in terms of readability, morphological structures, lexical choices, syntactic patterns, and semantic features. Using Open Brain AI, the study analyses 10 essays generated by the two models, forming a corpus of ~10,000 words. Readability analysis showed that ChatGPT uses longer sentences than DeepSeek. While DeepSeek produces texts suitable for high school students, ChatGPT's texts are more appropriate for college students. Stylometric analysis indicates that ChatGPT demonstrates a higher density in adjectives, adpositions, auxiliary verbs, subordinating conjunctions, and verbs, while DeepSeek shows a higher density in adverbs, coordinating conjunctions, nouns, particles, and pronouns. In addition, ChatGPT-generated texts exhibit richer and more diverse lexicons. Additionally, syntactic analysis showed that ChatGPT leads to the use of adverbial modifiers and direct objects. By contrast, DeepSeek leads determiners, nominal subjects, objects of prepositions, and roots. Finally, the findings reveal that DeepSeek outperforms ChatGPT in incorporating semantic features, except when events and ordinal and cardinal numbers are used. These insights highlight the distinctive characteristics of each AI tool, which can inform strategies for enhancing writing style.

ARTICLE HISTORY

Received 23 July 2025

Revised 21 August 2025

Accepted 22 August 2025

KEYWORDS

AI-generated texts;
comparative study;
ChatGPT; DeepSeek;
stylometric analysis

SUBJECTS

Cognitive Science; Sign
Language, Braille, and
Other Linguistic
Communication; English
Language; Morphology;
Phonetics and Phonology;
Psycholinguistics

Introduction

The rapid advancement of Artificial Intelligence (AI) tools has revolutionized the field of natural language processing (NLP), enabling machines to generate texts that increasingly resemble human writing (Alqahtani et al., 2023). Among the most prominent AI language models are ChatGPT and DeepSeek, both of which have demonstrated remarkable capabilities in producing coherent, contextually relevant, and stylistically diverse texts (Kerimbayev et al., 2024). Many previous studies have compared AI-generated texts (AIGTs) with human-authored texts (HATs) (e.g. Ma et al., 2023; Sardinha, 2024; Skowron & Bączkowska, 2023). AIGTs have also been explored using distinct text features (Opara et al., 2024). Some studies have reported that the high quality of AIGTs makes it difficult for readers to differentiate between HATs and AIGTs (Ma et al., 2023). In contrast, variations in AIGTs and HATs are phonological, syntactic, morphological, and lexical (Georgiou, 2024). However, the stylometric features of AIGTs have not been explored (Sardinha, 2024). This brings a niche to the present study.

Furthermore, some previous studies have focused on the stylometric differences between HATs and AIGTs (Opara et al., 2024; Sardinha, 2024; Shah et al., 2023); however, the stylometric features of different AI models have not been explored. In this study, we differed from previous studies by analyzing the stylometric features between two texts generated by AI tools. Moreover, existing research has used models, such as BERT and DeBERTa to process the stylistic features of written texts (Alshammari & Elleithy, 2024; Goswami et al. 2024). We adopted Open Brain AI to process and calculate the differences between the two AIGTs. Therefore, we strike such a research gap by comparing the stylometric features of two AI

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tools, i.e. ChatGPT and DeepSeek, to identify the similarities and differences between these two AI tools in generating texts.

As AIGTs become increasingly integrated into various domains, from creative writing to technical documentation, understanding these stylistic differences is crucial for optimizing their applications and ensuring alignment with user expectations (Alharthi, 2025). Additionally, previous studies have focused on the parametric features of texts, such as syllable count, pronunciation ratio, and word length (Shah et al. 2023). We examined linguistic features, such as syntax, lexicon, semantics, and readability. In this study, we aim to unveil the nuances that differentiate these models and shed light on their respective strengths and limitations. Hence, this study seeks to contribute to the broader discourse on AI-generated content by providing insights into how stylistic variations affect readability, engagement, and perceived authenticity. This study explored the following questions:

1. How do DeepSeek and ChatGPT differ in generating readable texts?
2. Do ChatGPT and DeepSeek generate the same stylometric features?

Literature review

Stylistics, as a discipline, examines the linguistic choices that shape the form and function of a text, including syntax, lexicon, tone, coherence, and rhetorical devices (Jaashan, 2014; Stockwell & Whiteley, 2014). With the advent of AI-powered language models, the study of stylometric features has expanded to encompass machine-generated texts, offering new opportunities to explore how computational systems emulate and diverge from human linguistic patterns.

Theoretical background

The stylometric features of AIGTs focus on some criteria related to the syntactic, lexical, semantic, and readability of texts. The existing studies differ in terms of their criteria, although their focus is relative. Kumarage et al. (2023) listed stylometric features based on three criteria: phraseology, punctuation, and linguistic diversity. The first stylometric feature, i.e. phraseology, refers to the organization of words and phrases to make a piece of writing. Punctuation is the second feature that represents the way in which authors use different punctuation marks. The third feature is linguistic diversity, which quantifies the authors' use of different words in writing in a way that helps the readability and richness of the text. Additionally, Banat's (2024) study focused on the stylometric fingerprint of ChatGPT translation from Arabic to English, using the same model to analyze the stylometric features of ChatGPT and DeepSeek-generated texts. Banat (2024) used four criteria to analyze the stylometric of texts, including phonology, vocabulary richness, sentence length, and stylistic features. Vocabulary richness was measured using the Type-Token Ratio (TTR) and Herdan's C. Sentence length was evaluated using the Flesch-Kincaid Grade Level and Gunning Fog Index. Punctuation usage was measured using the frequency of commas, periods, and other markers. Stylistic features were evaluated using part-of-speech (PoS) and named entity recognition (NER). Herrmann et al. (2021) focused on computational stylistic methods that employ metrics, such as lexical diversity, sentence complexity, and sentiment analysis to quantify stylistic differences, enabling systematic comparisons between texts. The features of texts concern researchers to obtain a clear-cut approach to analyzing AIGTs. Long Short-Term Memory (LSTM) is a deep learning model characterized by high accuracy metrics exceeding 95% used in analyzing texts (Chimata et al., 2024; Goswami et al. 2024). Other pretraining models for text analysis include BERT and DePERTa (Chimata et al., 2024). Researchers have also employed pre-trained models on large datasets to enhance classification accuracy (Goswami et al. 2024).

Stylometric analysis

Stylometric analysis aims to identify the characteristics of written discourse to highlight writers' styles (Iqbal et al., 2023). Relying on computational techniques, stylometry uses methods, such as examining word

sequences (n-gram), lexical frequency (word occurrences), sentence structure, and recognizing patterns using some learning techniques (Banat, 2024). Some researchers have identified three categories required for stylometric analysis: phraseology, punctuation marks, and linguistic diversity (Kumarage et al., 2023).

Recent research has focused on identifying and quantifying stylistic features of AIGTs, such as tone, readability, and lexical richness. For example, studies have used sentiment analysis and readability indices to compare the stylistic outputs of different models (Alafnan & MohdZuki, 2025; Dale & Viethen, 2021). Other studies have explored how AI models handle figurative language, humor, and cultural references, revealing both their capabilities and limitations in replicating human stylistic creativity (Ippolito et al., 2022; Zellers et al. 2020). These findings highlight the need for a systematic and data-driven approach to stylistic analysis, particularly in comparative studies.

Early work in computational stylistics focused on authorship attribution and stylistic fingerprinting using metrics, such as lexical diversity, syntactic complexity, and n-gram frequency (Juola, 2007; Stamatatos, 2009). More recently, computational stylistics have been extended to (Stamatatos, 2009) AI-generated texts, with researchers examining how language models replicate or diverge from human-stylistic conventions (Baron, et al. 2009; Grieve et al., 2010). Studies have shown that AI models can exhibit distinct stylistic traits based on their training data and architectural design, thus making computational stylistics a valuable tool for comparative analysis.

Large language models: ChatGPT and DeepSeek

The development of transformer-based language models, such as GPT-3 and its successors, has revolutionized NLG by enabling the production of coherent and contextually relevant texts. These models rely on self-attention mechanisms and large-scale pretraining to capture linguistic patterns, allowing them to generate text that mimics human writing. However, research has highlighted variations in the stylistic outputs of different models influenced by factors, such as training data diversity, fine-tuning strategies, and model size (Radford et al. 2019; Raffel et al., 2020). For instance, ChatGPT, optimized for conversational applications, often prioritizes fluency and adaptability, whereas models, such as DeepSeek may exhibit specialized stylistic features tailored to specific tasks or domains. AI language models, such as OpenAI's ChatGPT and DeepSeek are built on transformer-based architectures that leverage self-attention mechanisms to process and generate text (Naveed et al., 2023). These models are trained on a vast corpus of human-written text, enabling them to learn syntactic structures, semantic relationships, and stylistic conventions (Zheng, 2024). However, despite their shared foundational principles, variations in training data, model sizes, fine-tuning techniques, and design objectives can lead to distinct stylistic outputs. For instance, ChatGPT, known for its conversational fluency and adaptability, often prioritizes coherence and contextuality (Bansal et al., 2024), whereas DeepSeek, with its specialized training and optimization (Liu et al., 2024), may exhibit unique stylistic traits tailored to specific applications or domains.

ChatGPT

ChatGPT has shown remarkable capabilities in generating texts across different domains and features a human-like tone (Sarode & Bhamare, 2023). ChatGPT surpasses many language models in performing certain tasks and outperforms many LLMs in zero-shot learning (Bang et al., 2023). These LLMs could potentially impact the development of academic writing and scholarly publications (Bhatia, 2023). Its accuracy is mounted to reach ~80% of its output in certain fields (Mitrović et al., 2023). In contrast, ChatGPT is not free from challenges and struggles. Mitrović et al. (2023) stated that ChatGPT produces outputs that lack specific details and hallucinations. The ChatGPT also lacks reasoning, with an average accuracy of 63.42% in some fields (Bang et al., 2023).

DeepSeek

DeepSeek, a cutting-edge AI model, specializes in detecting context-aware errors and excels in technical tasks, such as coding and logical reasoning. This offers a cost-efficient open-source solution. It also stands

out in generating engaging and creative content through dynamic storytelling and enhancing learning experiences by making them more immersive and enjoyable (Mohammed et al., 2025). Albuhairey and Algaraady (2025) argued that DeepSeek is among the recently emerged NLP models that have the ability to generate texts to analyze the semantic features of texts and are capable of modeling language in various contexts.

Previous studies

Empirical studies have explored the various levels of AIGTs. Skowron and Bączkowska (2023) examined the lexical cohesion quality of AIGTs and compared the quality of various GPT models. This study used an NPL-based text analysis to examine lexical cohesion. This study indicated similarities between AIGTs and HATs. Furthermore, the study averted that no single model produced high-quality text cohesion. Ma et al. (2023) compared AIGTs, specifically scientific texts and HATs, at the syntactic, semantic, and pragmatic levels. This study developed an analytical model based on human assessment. The findings showed that AIGTs feature factual errors. This study reported a gap in the written style between AIGTs and HATs.

Comparative studies on AI language models have explored their performance across various dimensions, including coherence, creativity, and stylistic consistency. For example, studies comparing GPT-3 and BERT have highlighted differences in their abilities to generate contextually appropriate and stylistically diverse texts (Liu et al., 2024; Zhang et al., 2022). Georgiou's (2024) study, which compared HATs and AIGTs, attained some linguistic components, such as phonological, readability, morphological, syntactic, and lexical features to represent the inclusion of these criteria in HATs and AITGs. Similarly, research on domain-specific models, such as those used in legal or medical text generation, has demonstrated that fine-tuning can shape stylistic outputs (Chalkidis et al., 2023; Johnson et al., 2023). These studies underscore the importance of examining stylistic differences to understand the strengths and limitations of AI models in various contexts. While existing research has made significant strides in understanding the stylistic features of AI-generated texts, a gap remains. We have not gone through a study that comprehensively compares the stylometric features of ChatGPT and DeepSeek.

AlAfnan (2025) compared DeepSeek and ChatGPT in text generation for composition and business communications. The findings revealed that ChatGPT shows better performance in terms of audience engagement, linguistic diversity, and persuasive content creation. The findings indicated that DeepSeek excels in grammatical accuracy, structural coherence, and factual consistency, making it ideal for formal reports and standardized business writing.

Additionally, lexical and cohesion analyses of academic texts generated by GPT models revealed high similarity to human-written content, with no single model consistently outperforming others across all linguistic indices (Skowron & Bączkowska, 2023). In summary, this research is informed by theories of NLG, which explore how AI systems produce texts that mimic human language. NLG theories emphasize the interplay between linguistic rules, probabilistic modeling, and contextual understanding, all of which influence the stylistic outputs of AI models. This study addresses this gap by conducting a systematic stylistic analysis of ChatGPT and DeepSeek and employing computational stylistics and NLG theories to compare their linguistic features.

Methods

Research design

We adopted a corpus analysis of two essays generated by ChatGPT 3.5 and DeepSeek-R1 on five tourist places on Socotra Island, Yemen; Alula, Saudi Arabia; Babylon, Iraq; the Pyramids, Egypt; and Petra, Jordan.

Data building

We instructed both ChatGPT and DeepSeek to write a 1000-word essay on each of the five tourist destinations: Socotra Island, Alula, Babylon, the Pyramids, and Petra. Each time we instructed these two AI

tools to write an essay on one of the above destinations with the following instructions, 'Write in English an essay on the tourist places in Socotra. The essay should orient tourists, and should contain precisely one thousand words and include an introduction, five body paragraphs, and conclusion'. They were then instructed to write about the second destination to obtain five essays written by each AI tool. Sometimes, we repeated the process several times until we obtained a precise number of words for each essay. We validated the corpus beyond the word count using word count statistics in Microsoft Word. The resulting essays from ChatGPT were compiled into one file, whereas those from DeepSeek were compiled into another. The corpus of this study consisted of 10 essays, totaling 10,000 words generated by ChatGPT and DeepSeek. The word count was selected based on specific research considerations to ensure comprehensive coverage and analysis.

These tourist destinations were selected as a topic for instructing AI tools to generate texts because these places are UNESCO-designated international heritage sites and for their widespread fame. Being famous landmarks, AI tools provide sufficient and detailed information to be incorporated into their tasks. As university professors, researchers may use AIGTs to teach academic writing to their students. Therefore, conducting a stylometric analysis aids instructors in identifying the focus on AI tools at various linguistic levels while analyzing students' writings and helping them develop and enhance their creativity in writing.

Data analysis

However, some applications, such as BERT and DePERTa (Chimata et al., 2024), have been utilized to analyze the stylometry of AIGTs. The study adopts Open Brain AI software to explore the morphological, lexical, syntactic, and semantic features and readability of the generated texts. The open Brain is an online software designed to offer linguistic assessment of texts (Themistocleous, 2024). This study follows Georgiou's (2024) framework for analyzing linguistic features and showing variations between HATs and AIGTs. However, this study emulates the stylometric features of two AI-based tools, DeepSeek and ChatGPT4. Five linguistic features are investigated in this study, i.e. morphological, lexical, syntactic, semantic, and readability features.

To compare the readability of DeepSeek and ChatGPT, we analyzed the Average Sentence Length (ASL), Flesch-Kincaid Grade Level (FKGL), Gunning Fog Index, and Automated Readability Index (ARI). On the other hand, four stylometric criteria were adopted (Figure 1). Frequencies and ratios were calculated to report the compression between the two AIGTs.

Morphological analysis	Lexical analysis	Syntactic analysis	Semantic analysis
↓ Adjective Adposition Adverb Auxiliary Coordinating conjunction Subordinating conjunction Noun Particle Pronoun Verb	↓ Characters Words Sentences Function Words (Total) Content Words (Total) Mean Sentence Length TTR Corrected TTR Herdan's C	↓ Adverbial modifier Adjectival modifier Conjunct Determiner Direct object Nominal subject Object of preposition Root	↓ Cardinal number Date Event Facility Geopolitical entity Location Nationalities or religious or political groups Ordinal number Organization Person Time Work of art

Figure 1. Categories of stylometric analysis.

Results and discussion

This study delves into reporting a readability comparison between DeepSeek and ChatGPT AIGTs. It also displays the stylistic features of these AIGTs at morphological, lexical, syntactic, and semantic levels.

RQ1: How do DeepSeek and ChatGPT differ in generating readable texts?

Revealing the readability difference between DeepSeek and ChatGPT occurs under four criteria. Average Sentence Length (ASL), Flesch-Kincaid Grade Level (FKGL), Gunning Fog Index, and Automated Readability Index (ARI).

Regarding ASL, [Table 1](#) shows that ChatGPT follows longer sentences than DeepSeek ChatGPT. DeepSeek used 26.06 words per sentence, while ChatGPT used 37.22 words per sentence. These ASL are higher than the normal ASL in academia, where 24.7 ASL were found by Anglophone authors (Deveci, 2019). These results can be interpreted as the tendency of these two AI tools towards complexity, which requires individuals with profound knowledge to understand their content. ChatGPT adopts ASL with 37.22, which is considered very long, with many subordinate and coordinate clauses. Content writers may benefit from DeepSeek in producing their essays better than ChatGPT, which obtains abnormal ASL. These findings partially contradict those of Akinwande et al. (2024), who reported that HATs feature higher total words than AIATs.

FKGL is the second criterion used; it refers to the grade level to comprehend the texts. [Table 1](#) shows that DeepSeek's gendered text score was 15.40, while the ChatGPT-generated text score was 20.36. However, the difference in FKGL scores between ChatGPT and DeepSeek, DeepSeek texts would imply that postgraduate students have advanced reading ability, while ChatGPT texts can be understood by high-level students, usually PhD candidates. These findings contrast with those of Han et al. (2023), who assessed the FKGL of ChatGPT-generated text and pointed out that ChatGPT maintains learner-friendly output across different languages.

The Gunning Fog Index (GFI) was the third criterion used for comparison. [Table 1](#) demonstrates that DeepSeek generated text scores of 16.40, while ChatGPT scored higher at 22.30. This indicates that the two AI tools reflect various readability levels. These findings align with those of Chen et al. (2024), who pointed out that ChatGPT uses higher GFI scores and long sentences. These findings can help users use DeepSeek in their writing due to their lower GFI scores. Hegazy (2024) revealed that AI-generated texts with low GFI offer students the opportunity to comprehend them.

ARI is concerned with the ability to read a piece of text by focusing on counting letters and sentences. DeepSeek employs 17.39, whereas ChatGPT scores are 23.62. These scores can be interpreted according to the U.S. grade level that DeepSeek can be read by twelfth-grade students, while ChatGPT text can be read by college students. These findings partially agree with Patel et al. (2024), who pointed out that language models, such as GPT-3.5 provide the most readable responses. This is consistent with the study by Lin-Zucker et al. (2025), which showed that ChatGPT can adapt its output based on readability constraints and is suitable for second-language learners.

RQ2: Do ChatGPT and DeepSeek generate the same stylistic features?

Answering this question requires analyzing the two generated texts at morphological, lexical, syntactic, and semantic levels.

Table 1. Average readability measures across the DeepSeek and ChatGPT.

Criteria	DeepSeek		ChatGPT	
Average Sentence Length (ASL)	26.06	NaN	37.22	NaN
Flesch-Kincaid Grade Level (FKGL)	15.40		20.36	
Gunning Fog Index	16.40		22.30	
Automated Readability Index (ARI)	17.39		23.62	

Morphological analysis

Table 2 shows that ChatGPT utilizes morphological features at a higher density than DeepSeek. ChatGPT uses adjectives, adpositions, auxiliary verbs, subordinating conjunctions, and verbs (640, 183, 316, 1342, and 640) more than DeepSeek (614, 121, 245, 1281, and 614), respectively. On the other hand, DeepSeek utilizes morphological features at a higher density in adverbs, coordinating conjunctions, nouns, particles, and pronouns (116, 130, 158, and 523) than ChatGPT (89, 43, 124, 181, and 522).

These findings align with those of Bansal et al. (2024), who highlighted ChatGPT's strong conversational fluency and adaptability, prioritizing coherence and contextuality. Meanwhile, as noted by Liu et al., DeepSeek may exhibit characteristics that make it suitable for specific applications or domains. The findings also agree with those of Albuhairey and Algaraady (2025), who conducted a comparative study assessing DeepSeek and ChatGPT in analyzing second language acquisition errors. The findings indicate that DeepSeek excels in grammatical precision and formal report writing, suggesting potential strengths in morphological analysis. However, ChatGPT was noted for its adaptability and audience awareness, which are valuable in educational settings. The findings also meet the idea of Dewanti et al. (2025), who stressed that any study involving English as a Foreign Language (EFL) learners revealed that students used ChatGPT to simplify, confirm, and elaborate on morphological and syntactic concepts.

Lexical analysis

The lexical analysis is shown in Table 3, focusing on the nine lexical features used by ChatGPT and DeepSeek. Starting with the number of words, Table 3 shows that both essays are relatively similar in length (DeepSeek = 5126, ChatGPT = 5079), although the specified number of words in the five written essays was 5000. This similarity in word count ensures that the analysis of the following features is performed on five essays with the same number of words:

DeepSeek wrote five essays with a 5000-words limit using 192 sentences, whereas ChatGPT wrote the same number of essays with the word limit using 134 sentences. This is reflected by the average (mean) sentence length, where DeepSeek's ASL was 26.06, slightly above the ASL typically used by humans, while ChatGPT's ASL was 37.37. These findings suggest that DeepSeek tends to be more user-friendly, whereas ChatGPT tends to produce more complex language, which may be challenging for common users to understand.

Table 2. Morphological analysis of DeepSeek and ChatGPT's AIGTs.

	Measure	DeepSeek		ChatGPT	
		Frequency	Ratio	Frequency	Ratio
1	Adjective	614	0.120	640	0.126
2	Adposition	121	0.024	183	0.036
3	Adverb	116	0.023	89	0.018
4	Auxiliary	245	0.048	316	0.062
5	Coordinating conjunction	53	0.010	43	0.008
6	Subordinating conjunction	1281	0.250	1342	0.264
7	Noun	130	0.025	124	0.024
8	Particle	158	0.031	181	0.036
9	Pronoun	523	0.102	522	0.103
10	Verb	614	0.120	640	0.126

Table 3. Lexical analysis of DeepSeek and ChatGPT's AIGTs.

	Measure	DeepSeek		ChatGPT	
		Frequency	Ratio	Frequency	Ratio
1	Characters	28,546	5.569	29,236	5.756
2	Words	5126	NaN	5079	NaN
3	Sentences	192	NaN	134	NaN
4	Function Words (Total)	1818	0.355	1864	0.367
5	Content Words (Total)	3186	0.622	3144	0.619
6	Mean Sentence Length	26.06	NaN	37.37	NaN
7	TTR	25.8	NaN	26.20	NaN
8	Corrected TTR	12.9	NaN	13.09	NaN
9	Herdan's C	0.841	NaN	0.843	NaN

DeepSeek employs function and content word ratios of 0.355, 0.622, respectively, whereas ChatGPT uses ratios of 0.367 and 0.619 for function and content words. Three features of lexical diversity were examined: Herdan's C, TTR, and Corrected TTR. In Herdan's C, the DeepSeek score was 0.841, whereas ChatGPT's score was 0.843. For Corrected TTR, DeepSeek scored 12.9, compared to ChatGPT's score of 13.09. DeepSeek scored 25.8 in TTR, while the ChatGPT's score was 26.20. Despite the minor differences between the two AI tools, the study found that ChatGPT exhibited a slightly higher lexical diversity than DeepSeek. Our findings can be interpreted in relation to those of HATS. Previous research findings have shown that HATs tend to obtain distinct words and lexical richness, whereas ChatGPT uses low lexical diversity (Giorgi et al., 2023; Reviriego et al., 2024). Furthermore, our findings are partially consistent with Reviriego et al. (2024), who stated that while both ChatGPT and DeepSeek generate coherent and appropriate texts, ChatGPT's lexical diversity reveals certain limitations in vocabulary variety. DeepSeek, which exhibits stylistic similarities to ChatGPT, likely shares comparable lexical characteristics, although dedicated studies are needed to confirm this.

Syntactic analysis

Table 4 shows the syntactic properties of the AIGTs produced by DeepSeek and ChatGPT. ChatGPT was found to obtain more adverbial modifiers, adjectival modifiers, and direct objects with frequencies (199, 659, 394, and 292), respectively, while these syntactic features occurred in the DeepSeek generated texts with frequencies (124, 658, 300, and 237), respectively. In contrast, DeepSeek surpasses ChatGPT in obtaining determiners, nominal subjects, objects of prepositions, and roots with frequencies (590, 267, 621, and 192), respectively, while ChatGPT obtains these four syntactic features with (526, 252, 19, and 134).

Overall, the two AI tools demonstrated similarities in two syntactic features: adjectival modifiers (DeepSeek scored 658, ChatGPT scored 659) and nominal subjects (DeepSeek scored 267, ChatGPT scored 252). They also exhibit distinct patterns in features, such as conjuncts (DeepSeek scores 300, ChatGPT scores 394) and objects of prepositions (DeepSeek scored 621, ChatGPT scored 19). These findings highlight the tendency of ChatGPT towards a descriptive style where descriptions and details are provided. DeepSeek tends to emphasize straightforward sentences and structural information spatially or temporally, owing to the extensive use of adverbs. This can be explained by AIAfnan (2025), who reported that ChatGPT outperforms DeepSeek in linguistic variation and audience awareness, while DeepSeek excels in grammatical precision and formal writing.

Semantic analysis

The researchers calculated Named Entity Recognition (NER) to analyze the semantic features of the generated texts between DeepSeek and ChatGPT. Table 5 shows that neither model includes NER related to time. DeepSeek included more NER in dates, facilities, geopolitical entities, locations, nationalities, political or religious groups, organizations, persons, and work of arts (56, 73, 111, 101, 39, 182, 99, and 19) than ChatGPT (27, 58, 57, 94, 31, 104, 75, and 5, respectively). In contrast, ChatGPT included more NER in cardinal numbers, events, and ordinal numbers (27, 9, and 6, respectively). These findings show that DeepSeek outperformed ChatGPT by including semantic features in the generated text. These findings

Table 4. Syntactic analysis of DeepSeek and ChatGPT's AIGTs.

	Measure	DeepSeek		ChatGPT	
		Frequency	Ratio	Frequency	Ratio
1	Adverbial modifier	124	0.024190	199	0.039181
2	Adjectival modifier	658	0.128	659	0.129750
3	Conjunct	300	0.058525	394	0.077574
4	Determiner	590	0.115099	526	0.103564
5	Direct object	237	0.046235	292	0.057492
6	Nominal subject	267	0.052087	252	0.049616
7	Object of preposition	621	0.121147	19	0.003741
8	Root	192	0.037456	134	0.026383

Table 5. Semantic analysis of DeepSeek and ChatGPT's AIGTs.

	Measure	DeepSeek		ChatGPT	
		Frequency	Ratio	Frequency	Ratio
1	Cardinal number	18	0.003512	27	0.005316
2	Date	56	0.010925	27	0.005316
3	Event	4	0.000780	9	0.001772
4	Facility	73	0.014241	58	0.011420
5	Geopolitical entity	111	0.021654	57	0.011223
6	Location	101	0.019703	94	0.018508
7	Nationalities or religious or political groups	39	0.007608	31	0.006104
8	Ordinal number	3	0.000585	6	0.001181
9	Organization	182	0.035505	104	0.020476
10	Person	99	0.019313	75	0.014767
11	Time	0	0.000000	0	0.000000
12	Work of art	19	0.003707	5	0.000984

agree with Albuhairey and Algaraady (2025) ideas, who pinpointed the ability of DeepSeek to generate texts and analyze the semantic features of texts.

Conclusion

Our study compared the readability and stylistic features of 10 essays generated by DeepSeek and ChatGPT. Our findings revealed that DeepSeek outperformed ChatGPT in several aspects, including readability rates. This finding is important for content writers to adopt DeepSeek while preparing notes for their study. College students may use DeepSeek to develop their writing style after finishing their essay writing themselves. EFL instructors can also use samples of DeepSeek's generated texts to illustrate some essay writing techniques to their students. Furthermore, DeepSeek also incorporates higher semantic features in its generated texts when compared to ChatGPT, thereby enhancing the clarity of meaning of DeepSeek's outputs. Users of AI tools can leverage such features to improve their content or even enhance their writing. A potential reason for the difference is DeepSeek's application of advanced algorithms designed to integrate semantic features, which significantly aid in understanding contexts.

Moreover, ChatGPT tends to focus on descriptions, as evidenced by its higher use of adjective modifiers, whereas DeepSeek tends to emphasize the temporal and special use of information owing to its frequent use of adverbs. In addition, ChatGPT-generated texts exhibit a slightly richer and more diverse lexicon than DeepSeek. This difference may be attributed to the training database used for each model. Notably, ChatGPT is trained on descriptive databases leading to richer vocabulary. In contrast, DeepSeek is trained on precision and clarity, leading to a straightforward use of language.

We learned from these findings that AI tools can generate texts. However, these texts are characterized by complexity. Using these tools required the users to be careful about inserting appropriate instruction regarding the audience who will read the texts. Furthermore, we have learned that every AI tool outperforms others in certain features and underperforms other AI tools in other features. Getting acquainted with the strengths and weaknesses of each AI tool enables users to use each tool in the appropriate contexts. These findings are important for individuals interested in using AI for writing content. These can effectively combine the strengths of each tool. Freshman college students could use DeepSeek to obtain feedback about their writings, or they could use DeepSeek's generated text as models to follow. In contrast, senior students may find ChatGPT useful for increasing lexical density in their writing. However, students' understanding to the writing styles and nuances of the English language may be hindered when they over depend on AI-tools. Their authenticity may also be impacted too.

These insights are also significant for EFL instructors who can integrate these tools into their teaching practices. This differentiation is crucial for optimizing communication effectiveness in educational contexts and professional writing. On the contrary, EFL instructors should be careful when assessing students' writing. EFL students should develop their writing skills independently rather than depending on AI tools.

Despite the originality of our study, there are some limitations. First, the study relied on stylistic analysis, which revealed qualitative findings, and future research studies should include both qualitative

and qualitative features. Furthermore, the corpus used in this study was narrow, and future research could build a larger corpus. Future studies are recommended to analyze the stylometric features of AIGTs in different genres, such as scientific, medical, and similar genres. Additionally, ongoing research into the capabilities of these AI tools is essential to further refine their applications and enhance user experience in various writing scenarios.

Author contributions

CRedit: **Hasan M. S. Jaashan**: Data curation, Funding acquisition, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing; **Wagdi Rashad Ali Bin-Hady**: Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The authors extend their appreciation to the Deanship of Scientific Research at King Khalid University for funding this work through the General Group Project (Under grant number GRP/37/46).

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Data availability statement

Data will be made available on reasonable request.

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